INTRODUCTION TO R

CSTAT

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AGENDA

1. What is R?

2. Why R?

3. How to use R?

I. Installation

II. Some Concepts and good practices

4. Hands on excises.

I. Data Preparation

II. Initial Data Analysis (Descriptive statistics)

WHAT IS R?

Statistical Programming Language

R is an interpreted programming language for statistical computing and graphics. Created by statisticians Ross Ihaka and Robert Gentleman and first shared in 1993. It considered to be successor of S language created by AT&T Bell laboratories.

The Comprehensive R Archive Network (**CRAN**) was founded in 1997 by Kurt Hornik and Fritz Leisch to host R's source code, executable files, documentation, and user-created packages.

WHY R?

Great research tool

1.Free! Every one can use it and every one can contribute.

2.Advanced statistical methods. Not available in other softwares.

3.Flexible. Simple calculations or complex data analysis report.

4.Powerful graphics. No need to manually edit figures.

5.Render multiple output formats. Can be interactive format.

6.Reproducible Research. Benefit the entire scientific research community.

… And many more.

HOW TO USE R?

Install R

Installing R

It is straightforward installing R on your machine. Follow these steps:

1. Go to the CRAN (Comprehensive R Archive Network) R website.
2. Choose to download R for Linux, Mac or Windows. Latest version: R-4.5.2 (released on 2025-10-31, [Not] Part in a Rumble).
3. For Windows users, just install 'base' and this will link you to the download file
4. For Mac users, make sure you download the right version for your CPU.

HOW TO USE R?

Install IDE: RStudio

Installing RStudio

R studio is the most popular integrated development environment (IDE) for R. It makes working with R easier, more productive, and organized, especially for new users.

Install steps:

1. Go to the RStudio download website (Posit).
2. Choose the 'Installers for your system' to download
3. Install RStudio as per the instructions.

HOW TO USE R?

Cloud options: Posit cloud

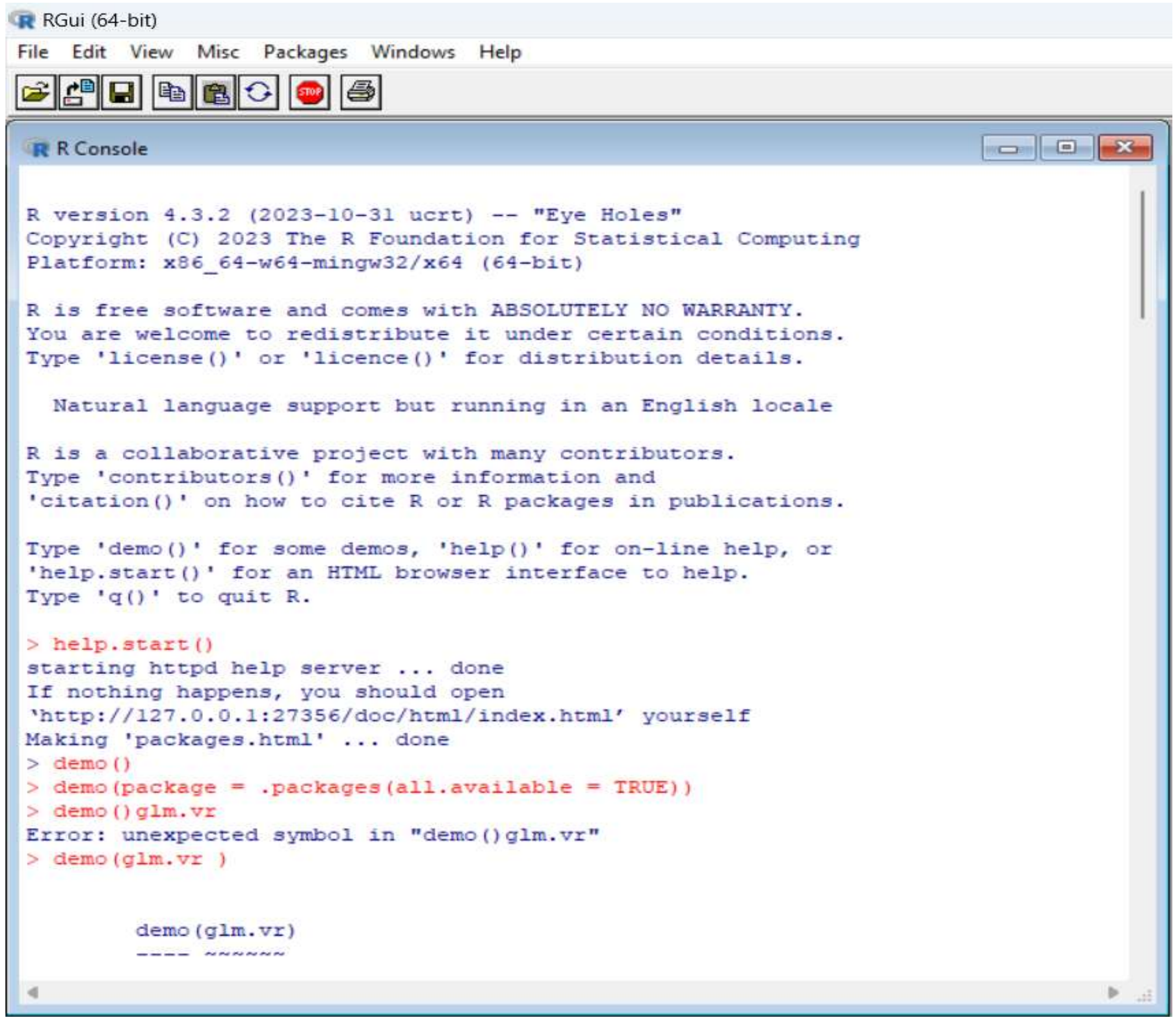
Posit cloud offers cloud solution to use R, Rstudio without installing on local machine.

Follow steps:

1. Go to website Posit cloud to register new user/login. Different plans with different limits. Free one limit 25 projects.
2. Create new workspace.
3. Click create new project.

HOW TO USE R?

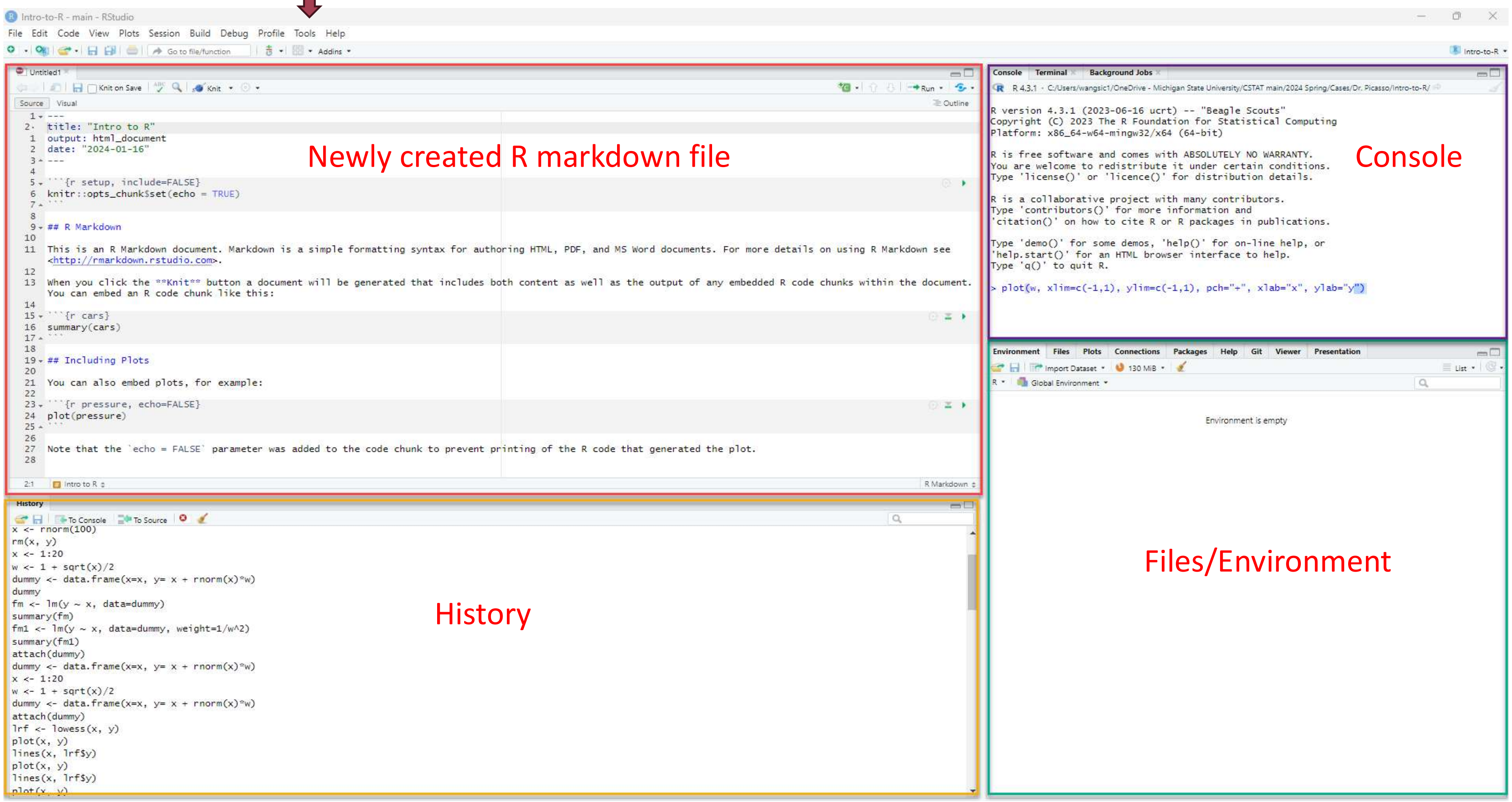
R GUI



HOW TO USE R?

R studio Layout

Tools>global options to change layout and settings



HOW TO USE R?

Some concepts

- **Packages**. a collection of functions, documentation, and data.

  install.packages(pkgnames), library(pkgnames)

- **Functions.** Get things done (outputs). Eg, mean(), sd(), inputs of functions

  called arguments

- **Object-oriented programming.** The output can be used as input,

Assignment: <- (alt+-)

- **Case sensitivity.** It treats great and Great differently. class() or typeof()
- **Data types.** numeric(real or decimal), integer, character, complex, and logical.
- **Data structures.** Vector, matrix, factor, list, data frame. class()

HOW TO USE R?

Good practices

- **Start with a clean environment.** Recommend to create a project, do not use setwd().
- **Consistent coding style**. Google's R Style Guide. Eg: name convention(not start with number), indentation, etc.
- **Comment the code.** Literate programming. code is explained. Comment starts with #
- **Use pipe |>.**
- **Use shortcut keys.** Tools → Keyboard Shortcuts Help or Alt + Shift + K.
  alt+enter: run code. Ctrl+alt+i:insert: chunk. Ctrl+shift+M: |>.
- **help(functionname) or ?functionname.** Get information of any specific named function
- **Github.** Version control is a time machine.
- **Rmarkdown or Quarto.** **Updates!!!!! and many more…**

RESOURCES

BOOKS

- R for Data Science by Hadley Wickham
- ggplot2: Elegant Graphics for Data Analysis by Hadley Wickham
- Advanced R by Hadley Wickham
- R Graphics Cookbook by Paul Teetor
- R in a Nutshell: A Desktop Quick Reference by Joseph Adler
- R in Action by Robert Kabacoff
- Introductory Statistics with R by Peter Dalgaard
- R for Everyone: Advanced Analytics and Graphics by Jared P. Lander

… and many more.

RESOURCES

WEBSITES

- An Introduction to R (https://cran.r-project.org/doc/manuals/R-intro.pdf)
- R-bloggers (https://www.r-bloggers.com/)
- R pub (https://rpubs.com/)
- R-seek (https://rseek.org/)
- Data carpentry lessons (https://software-carpentry.org/lessons/)
- Quick-R (https://www.statmethods.net/)
- MSU R Programming for Data Sciences course online book (jeffdoser.com)
- The R graph gallery (https://r-graph-gallery.com/)
- Good practices (https://swcarpentry.github.io/good-enough-practices-in-scientific-computing/)

HANDS-ON

DEMOS

DATA PREPARATION

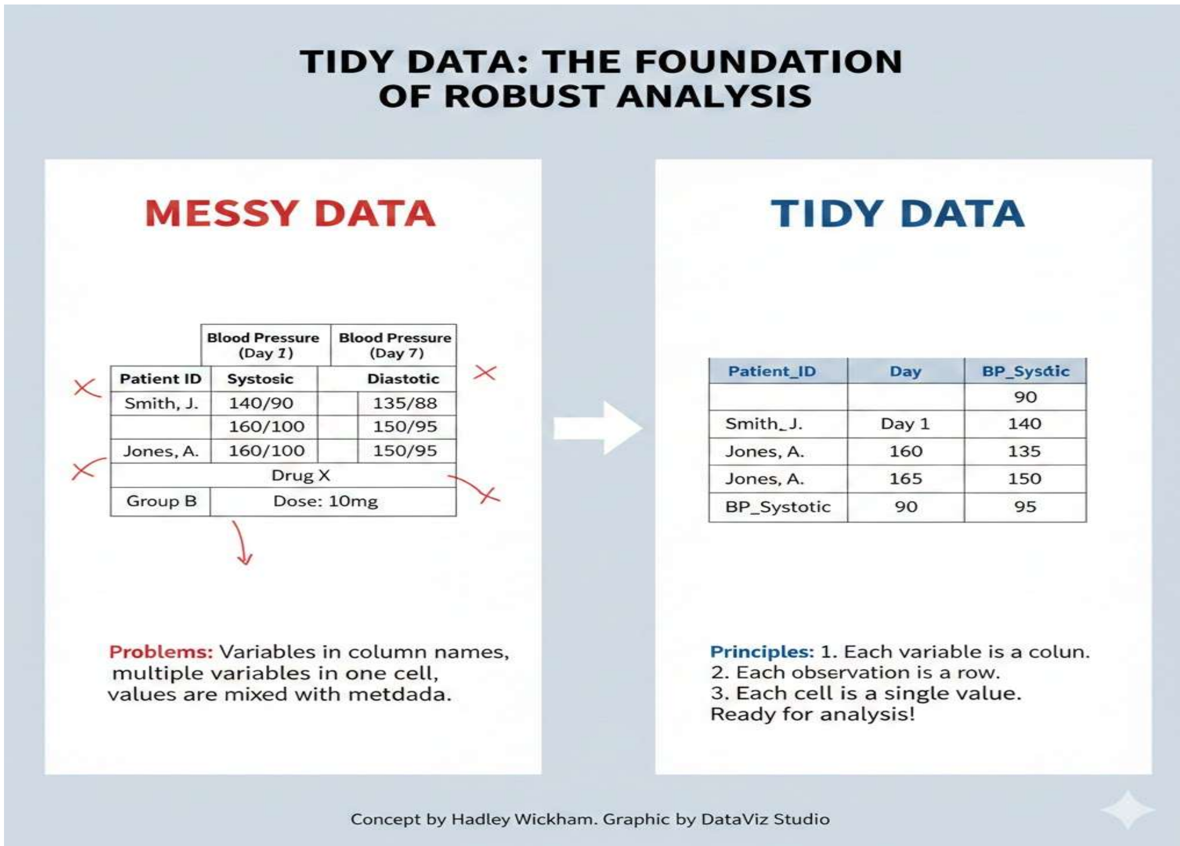
This class assume we all have clean data, which means the data is ready for analysis.

- Do not underestimate the time for data preparation. Often than not most of the time will be spent here.
- Tidy data concept. Please refer to Wickham 2014. 3 Features of **tidy data**:
  1. Each variable form a column.
  2. Each observation forms a row
  3. Each type of observational unit forms a table.

- package tidyverse is a great help.

References: Wickham, Hadley. 2014. "Tidy Data". *Journal of Statistical Software* 59 (10):1-23. https://doi.org/10.18637/jss.v059.i10.

DATA PREPARATION



INITIAL DATA ANALYSIS (IDA)

IDA primarily ensures transparency and integrity of preconditions to conduct appropriate statistical analyses in a responsible manner to answer predefined research questions [1]. It includes metadata setup, **data cleaning**, data screening, Documentation and initial data reporting, Refining and updating the statistical analysis plan [2].

Data screening is related to descriptive statistics. It includes:

1. Distribution of single variables.
2. Missing data.
3. Association between variables.

References:

[1]. Baillie M, le Cessie S, Schmidt CO, Lusa L, Huebner M, Topic Group ”Initial Data Analysis” of the STRATOS Initiative. Ten simple rules for initial data analysis. PLoS Computational Biology. 2022;18(2):e1009819
[2]. Huebner, M., le Cessie, S., Schmidt, C.O., & Vach, W. (2021). A Contemporary Conceptual Framework for Initial Data Analysis. Observational Studies, 4, 171 - 192.

INITIAL DATA ANALYSIS (IDA)

```
remotes::install_github("higgi13425/medicaldata")
```

Acute Bacterial Meningitis Dataset:

I will use ABM dataset to demonstrate descriptive data analysis. Please use own clean dataset to follow along.

We will move to rstudio workspace.

References: Spanos A, Harrell FE Jr, Durack DT. Differential diagnosis of acute meningitis. An analysis of the predictive value of initial observations. JAMA. 1989 Nov 17;262(19):2700-7. PMID: 2810603.